Answers

Chapter 1

1 (a) Pressure on keys, display of calculated value, (b) electrical signal, sound, (c) radio signals, sound
2 Open – no feedback, closed – feedback, see Section 1.3.1 and 1.3.2
3 (a) Measurement – temperature sensor, controller – thermostat, correction – heater, process – water bath, (b) measurement – rotary speed sensor, controller – motor, process – shaft, (c) measurement – sensor of thickness, e.g. LVDT, controller – differential amplifier, correction – rollers, process – steel strip
4 See Fig. A.1
5 Measurement – thermistor with resistance-to-voltage converter, comparison – differential amplifier, correction - relay and heater, process – the enclosure being controlled; measurement – level probe, controller – relay and solenoid valve, correction – flow control valve, process – water tank, measurement – level probe
6 (a) $x - y$, (b) $x + y$, (c) $x - y - z$
7 Eg. (a) resistance temperature detector, (b) limit switch, (c) LVDT, (d) tachogenerator
8 (a) Push button activate, spring return: initially output 2 is at pressure, when button pressed output 2 is vented, (b) solenoid activated, spring return: initially output 2 is vented, when button pressed output 2 is pressurised,
9 (a) Pressing causes piston to move to left, release gives return, (b) pressing causes piston to move to left, release gives movement to right
10 See Figure A.2
11 (a) See Figure A.3(a), (b) see Figure A.3(b)
12 Ball operates a lever which opens or shuts a valve, depending on the height to which the ball is floating. The comparison element is the position of the lever, the control element is the lever, the correction element the valve, the process the water in the cistern and the measurement is the floating ball.

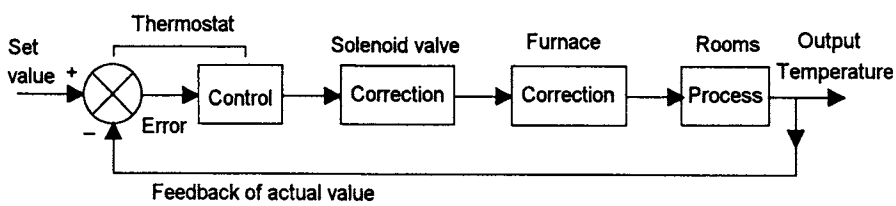

Figure A.1 *Chapter 1, problem 4*

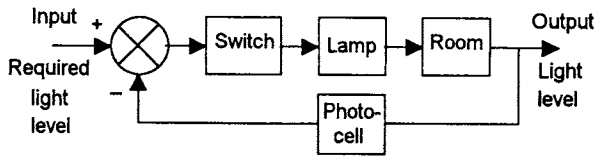

Figure A.2 *Chapter 1, problem 10*

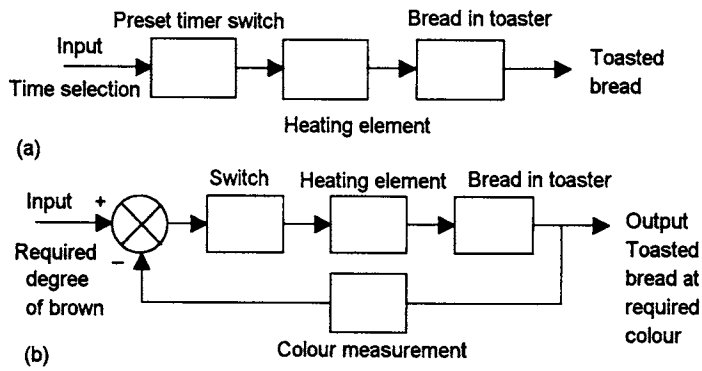

Figure A.3 *Chapter 1, problem 11*

Chapter 2

1 10 V
2 100
3 (a) $k_1x = M\frac{d^2y}{dt^2} + c\frac{dy}{dt} + (k_1 + k_2)y$,
(b) $T = I\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + k\theta$,
(c) $v = L\frac{di}{dt} + Ri$,
(d) $p = c\frac{d\theta}{dt} + cpq\theta$, where $\theta = \theta_o - \theta_i$
4 $5/[s(s + 1)]$
5 1.7
6 $2/(s + 1.2)$
7 $1/(1 + RCs)$
8 (a) $G_1G_2G_3$, (b) $G_1G_2/(1 + G_1G_2G_3)$, (c) $G_1G_2G_3/(1 + G_1G_2G_3)$, (d) $G_1G_2/(1 + G_1G_2H_1 + G_2H_2)$
9 (a) $1/[s(s + 2) + 2]$, (b) $K/[s(s - 1) + K(s + 2) + Ks]$, (c) $2/(s^2 + 9s + 6)$
10 $K_1N/[s(sL + R) + K_1K_2N]$
11 $75/[(s + 1)^2(s + 2) + 7.5]$
12 $Y(s) = [2/(s^2 + 4)]X_1(s) + [s/(s^2 + 4)]X_2(s)$
13 $Y(s) = [4/(s^2 + 16s + 40]X_1(s) + [s/(s^2 + 16s + 40]X_2(s)$

Chapter 3

1 $3/s$
2 2
3 $5/s^2$
4 e^{-5t} V

5 (a) $(5/3)(1 - e^{-3t})$ V, (b) $5e^{-3t}$ V
6 (a) $6(1 - e^{-t})$ V, (b) $6e^{-t}$ V
7 (a) $2(1 - e^{-2t})$ V, (b) $\frac{1}{2}[t - \frac{1}{2}(1 - e^{-2t})]$ V
8 (a) $5/(s - 1) - 4/(s - 2)$, (b) $4/(s + 1) - 3/(s + 2)$, (c) $2/(s + 1) - 3/(s + 1)^2$
9 $24 - 12e^{-2t} - 4e^{-4t}$
10 $8e^{-2t} - 8t\,e^{-2t}$
11 (a) Critical, (b) overdamped, (c) underdamped
12 $-1.5 + 3.0t + 1.5e^{-2t}$
13 $0.5 - e^{-t} + 0.5e^{-2t}$
14 $0.5(e^{-t} - e^{-3t})$
15 $0.5(1 - e^{-10t})$
16 10, 0.05
17 Underdamped
18 Critically damped
19 1/53 s

Chapter 4

1 (a) 9.2 s, 8.8 s, (b) 2.3 s, 2.2 s, (c) 0.77 s, 0.73 s
2 69.1 s, 65.9 s
3 8.7 s, 20 s, 19.1 s
4 (a) 10 rad/s, 0.2, 0.16 s, 53%, 2 s, (b) 7 rad/s, 0.29, 0.23 s, 39%, 2.0 s
5 4 rad/s, 0.63, 0.51 s, 7.8%, 1.6 s
6 2.6
7 Stable (a) and (c); oscillatory (b) and (d)
8 (a) Stable, (b) unstable, (c) unstable
9 (a) Stable, (b) unstable, (c) stable
10 (a) Stable, (b) unstable

Chapter 5

1 (a) 2, 90°, (b) 2, 180°, (c) 2.2, 26.6°
2 (a) $3/(5 + j\omega)$, (b) $7/(2 + j\omega)$, (c) $1/(10 + j\omega)(2 + j\omega)$
3 (a) 1.34, −26.6°, (b) 1.06, −45°
4 See Figure A.4
5 See Figure A.5

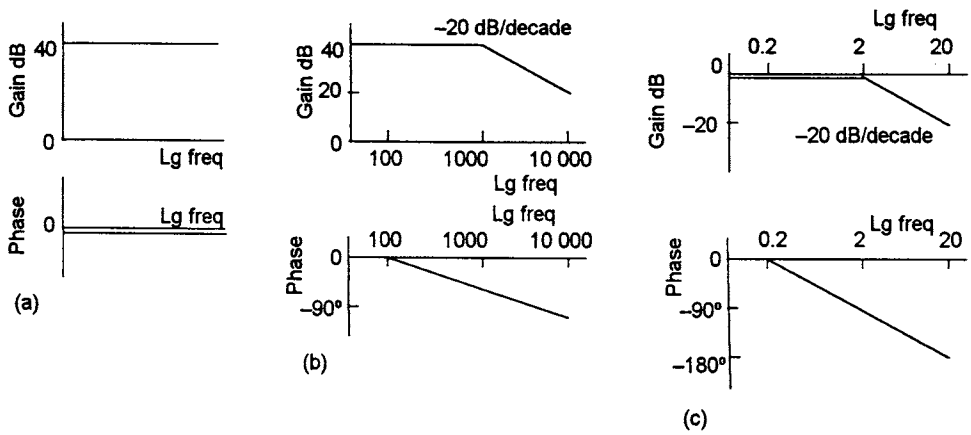

Figure A.4 *Chapter 5, problem 4*

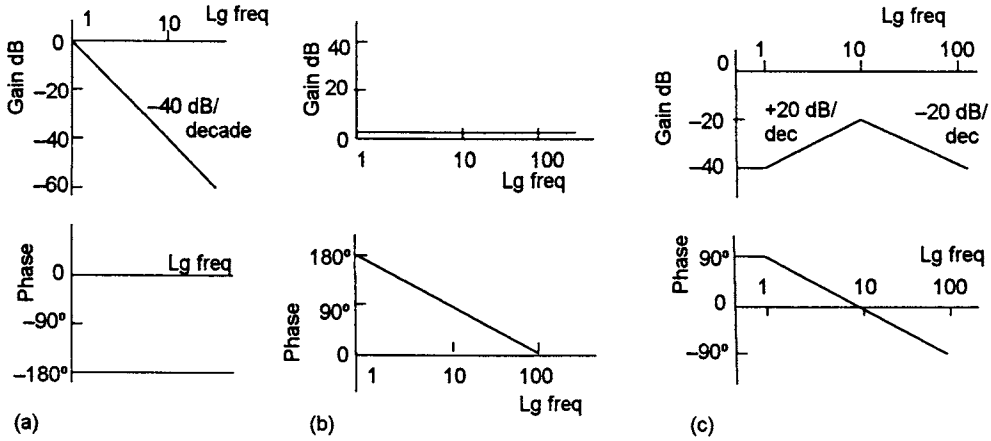

Figure A.5 *Chapter 5, problem 5*

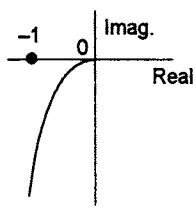

Figure A.6 *Chapter 6, problem 1*

6 (a) $1/s$, (b) $3.2/(1 + s)$, (c) $2/(s^2 + 2\zeta s + 1)$, (d) $3.2 \times 100/(s^2 + 2\zeta s + 100)$, (e) $10/(s^2 + 0.8s + 4)$
7 $16 \times 15/(s + 15)$
8 1.82, 50°
9 (a) Stable, (b) about 36 dB, (c) about 80°
10 (a) Stable, (b) about 15 dB, (c) about 43°
11 (a) Stable, (b) about 15 dB, (c) about 32°
12 About 5.7 dB, reduced by factor of 1.9
13 About 34°, 1.7 Db, reduced by factor of 1.2
14 55°
15 $(0.38s + 1)/(0.13s + 1)$
16 $(10s + 10)/(80s + 1)$

Chapter 6

1 See Figure A.6
2 (−1, j0) point not to be enclosed
3 8.0 dB, 20°
4 4.4 dB, 10°
5 3.5 dB, 11°
6 28 dB, 76°

Chapter 7

1 (a) −10°C to +30°C, (b) 40°C
2 (a) 2°C, (b) 10%
3 0.40 mA/°C
4 (a) 255 rev/min, (b) 324 rev/min
5 50%, 60%
6 1 unit
7 3990
8 See Figure A.7
9 $K_p = 2$, $T_i = 5.5$ min, $T_d = 1.4$ min
10 $K_p = 1.3$, $T_i = 6$ min, $T_d = 1.5$ min
11 $K_p = 0.62$, $T_i = 3$ min, $T_d = 0.75$ min

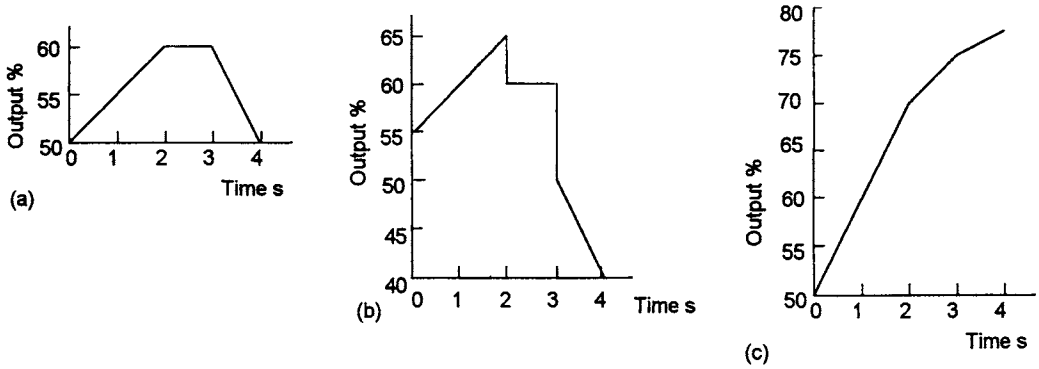

Figure A.7 *Chapter 7, problem 8*

12 $\dfrac{10(K_d s^2 + K_p s + K_i)}{s(s=5)(s+10) + 10(K_d s^2 + K_p s + K_i)}$

13 (a) –40°, (b) 15°

14 (a) $4 - 4\,e^{-0.5t}$, (b) $5[1 - 0.997\,e^{-0.05t}\sin(0.63t - 1.49)]$